

Software Requirements Specification for

HR Management System

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Abstract

HR Management System is a Web-based application that automates and streamlines an organization’s Human Resource activities. HR traditionally performed activities, such as records management, attendance tracking, leave processing, and payroll, manually; therefore, there were multiple ways in which an organisation could become inefficient, inaccurate, and have delays due to operational issues and human error. The HR Management System addresses these issues by providing a single platform that combines security, simplicity, and functionality for all HR and employee-related information.

The HR Management System will serve as a centralized point of access for a number of key administrative functions for the HR Department of a company. This includes allowing employees to log in to view and update their personal information, submit leave requests and check their attendance history, and view their salary information if applicable. The HR Department administrator will have the ability to create, modify, or delete an employee record as well as approve or deny an employee leave request and record/update employee attendance. The administrator will also be responsible for managing the payroll of employees. An employee's ability to access this information will depend upon the employee's role and permission levels, and they will only be able to view those records for which they have a right to access. To help ensure the integrity and security of the HR Management System, the data will be validated.

This document describes the Software Requirements Specification, outlining completely the Functional Specifications and Non-Functional Specifications of the software architecture and satisfies the overall purpose, objectives and goals for implementing a digital solution; this document will include a general discussion of the development processes and methods used, any associated risks and the resulting system diagrams. This project outlines the process of developing a digital solution as a way for organisations to maximise the efficiency of their HR departments, improve the transparency of information within their operations, and improve overall HR department effectiveness in the modern workplace.

Introduction

The purpose of the HR Management Systems is to automate and digitise the administrative Human Resources duties within a company. As companies grow, it becomes either impossible or extremely labor-intensive to administer the employee records manually. The HR Management System provides an ideal solution to this issue, providing a consistent and methodical way to perform HR functions while enhancing the efficiency, speed, accuracy, and dependability of HR management. The introduction will contain the purpose and reason for the development of the HR Management Systems, as well as the issues that it solves and the objectives it will accomplish.

1.1 Background

The HR Management office is a fundamental area of supervision in any organization. HR Management offices maintain employee records, process leave requests, track employees' attendance, provide compensation information to staff, and enable communications between companies and employees.

In addition, many organizations continue to operate HR Management offices manually or semi-automatically. These practices result in several challenges, such as inefficient processing, inaccurate employee records, delayed processing of decisions, and greater potential for error.

The need for businesses to transition from paper and spreadsheet methods of HR Management to digital services is critical in today's increasingly complex work environment. By implementing an online (web-based) HR Management System that consolidates all employee records and processes all requests electronically, companies will improve their efficiency in managing their workforce by eliminating the need for approval of requests through manual methods.

The Digital HR System is a centralized application that will simplify the development of modular applications to support employees and employers with a secure means to access sensitive information.

1.2 Motivation

Several factors motivated the development of this system:

1. Growing Demand for Human Resources

As an organization continues to grow and evolve, so do its human resources responsibilities. Additionally, organizations that rely on manual record-keeping will continue to experience slower, repetitive, and error-prone processes.

2. Speedier Decision Making

HR managers and other managers involved with human resources require timely access to reliable, secure information to make sound decisions related to employee leave requests, salary increases, and employee evaluations.

3. Limited Access to Employee Data

Many employees do not have easy access to their own personal data (i.e., leave status, attendance history, pay history). An automated HR system can relieve employees from relying on HR staff to access this important information.

4. Availability of Automation

Because many human resources functions and processes are based on established procedures and workflows, they are suitable candidates for automation through a software solution.

5. Application of Theoretical and Practical Knowledge

The development of this project will allow me to apply my theoretical knowledge of web technologies, software engineering, and database management to an actual project in the real world.

1.3 Problem Statement

Manual systems and/or outdated workflow processes are still commonly employed to manage human resource activities by many organizations. These manual administrative processes result in multiple challenges, including the following:

1. The problems are maintaining current and accurate employee records.
2. The labour intensity of the leave approval process. [3]
3. A mistake was made in attendance records and payroll calculations. [3]
4. The absence of a central location for data storage increases the chances of duplicate/erroneous entries or loss of data.
5. Poor communication between employees and human resource departments[4].
6. Delay in creating salary and other human resource report documents.
7. Threat to the security of sensitive information due to the mismanagement of information

Not having an automated human resource management tool in place increases the risk of error through automation while decreasing the level of administrative burden. Thus, it is important to have a secure and centrally-managed automated system to simplify employee management as well as automate repetitive tasks and take away the potential for human error, whilst also providing timely and accurate communications between the employee and the human resources manager.

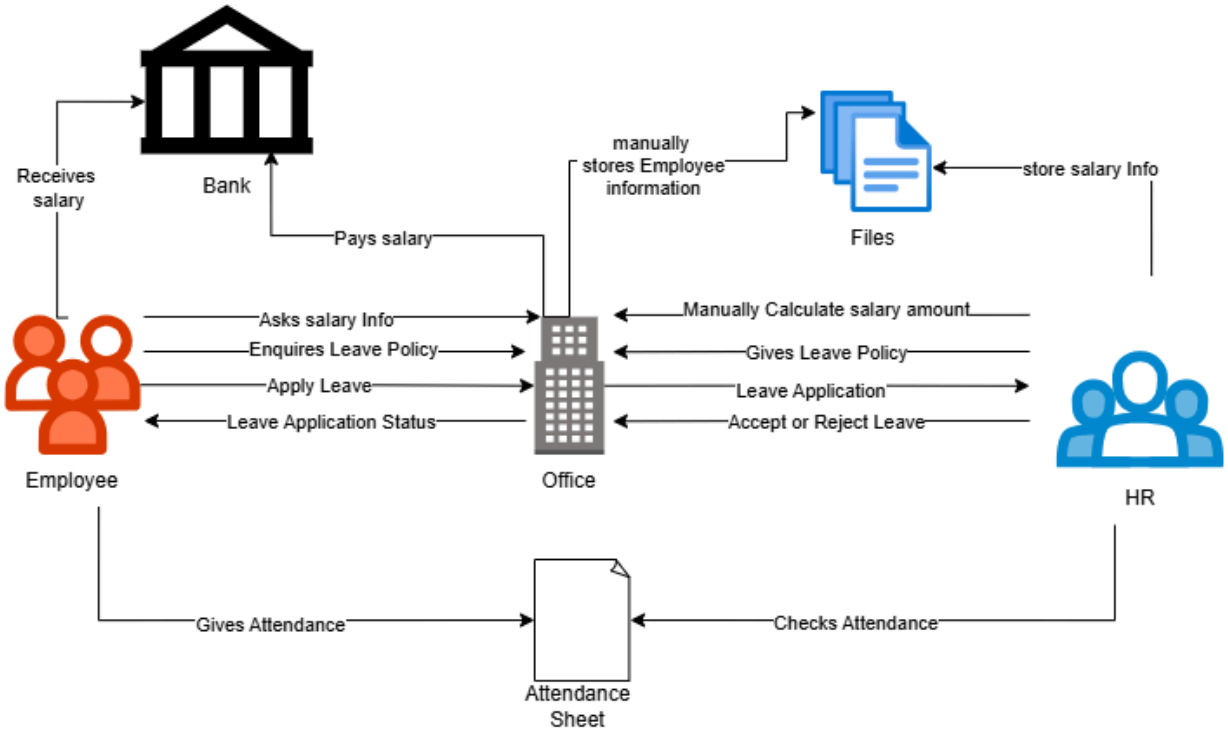


Fig 1: Existing System Rich Picture

1.4 Objective

The main objectives of the HR Management System are:

1. Automate employee management: Create a centralized, automated way to maintain employee records and not require any manual paperwork to update an employee's information.

2. Provide an easier way to apply for and approve or decline leave requests: Allow employees to easily submit the leave request, and allow the HR department to approve or decline those requests.

3. Maintain accurate attendance records: Provide HR the ability to enter attendance records into the system, while also allowing employees access to view their own attendance history.

4. Automate payroll calculations and payments: Automate the calculation of an employee's salary depending on attendance, deductions, and allowances, providing HR with a way to pay an employee quickly and easily.

5. Make attendance, leave status, and salary information easily accessible to employees: An employee can easily see his or her own information through a user-friendly interface.

6. Ensure the protection of sensitive employee data: Provides the ability for the secure handling of sensitive employee data by utilizing authentication, authorization, and encryption methodologies.

7. Create a structure that will allow for the future growth and long-term usefulness of the system: Create the system architecture that will allow for future expansion by adding features such as analytics, notifications, or biometry.

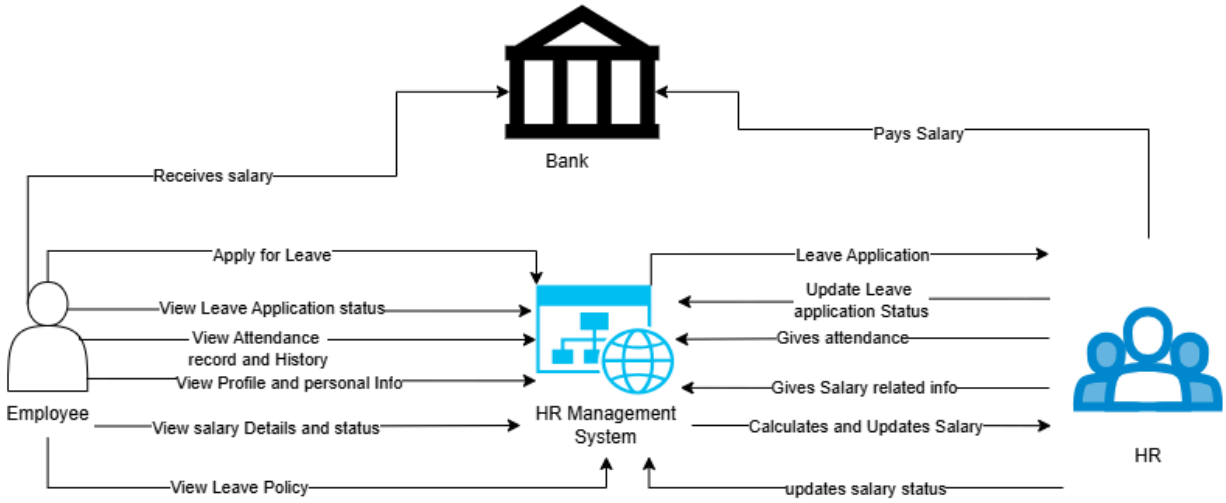


Fig 2: Proposed System Rich Picture

System Overview

The HR Management System allows organizations to automate and streamline their HR functions quickly and efficiently. In the past, HR functions required a lot of manual effort (such as record keeping), as well as multiple paper-based systems that involved many levels of redundancy. As a result, many HR functions were done outside of normal work hours, or there were considerable delays between the time that an employee was hired and worked for an employer.

The HR Management System provides a single location for storing, maintaining, and processing employee information, which is accessible to employees and HR staff on time. Both employees and HR staff have access to this information on an as-needed basis through a user-friendly interface.

The HR Management System provides a secure, scalable storage space for sensitive information and permits organizations to provide HR services that facilitate their long-term growth.

2.1 System Users

The system identifies two primary user groups. Each user is provided with specific functionalities based on their role and permission level.

1. Employee

Users of the System will consist of Employees. The role and responsibilities of Employees when using the System is as follows, but not limited to:

a) Creating an Employee System account on the System/Virgo platform.
b) Logging into the System/Virgo System.
c) Updating and changing an Employee's profile.

d) Applying for and tracking Leave of Absences (LOA's) for Work and viewing the status of those applications.
e) Viewing the Work, Holiday and Sick Leave Policies.
f) Viewing Attendance History every week and month.
g) Tracking Salary, Deductions and Payment status.

Employees will only have access to their individual record(s) within the System. All Leave of Absences, application submissions, and requests for HR must be approved by HR prior to an Employee being able to submit their applications.

2. HR

HR Administrators run the operations of the System. Responsibilities Include:

a) Viewing a list of all active employees.
b) Creating and maintaining employee Records, including verifying all Change Requests that come from Employees.
c) Reviewing and approving/denying Employees' Leave Applications.
d) Maintaining Attendance Records for Employees.
e) Processing Salary Allocations and payments.
f) Generating Basic HR Reporting Based on Stored Data.

Users of the HR System make decisions and perform the necessary functions to support and manage an organization.

2.2 System Architecture

This architecture ensures smooth operation, secure communication, and strong separation of responsibilities. The HR Management Systems employs a 3-tier architecture, which includes:

1. Presentation Layer

Frontend is users view the HR Management System in a web browser and interact with the dashboards, forms, and reports that are generated. The HR Management System tracks user interactions.

2. Application Layer

The user authorisation and authentication are carried out by the application layer. It offers features for calculating salaries, validating leaves, and confirming attendance. In order to prevent one user from accessing another's dashboard, the application layer also controls access to user dashboards through role-based constraints.

3. Database Layer

To guarantee data confidentiality, correctness, and consistency, all data is kept in MYSQL-structured tables. Controlled and secure enquiries are used to access data for creation, retrieval, updating, and deletion as well as to give management the ability to make decisions. The HR Management System runs smoothly thanks to its architecture, which also offers secure communications and a distinct division of duties.

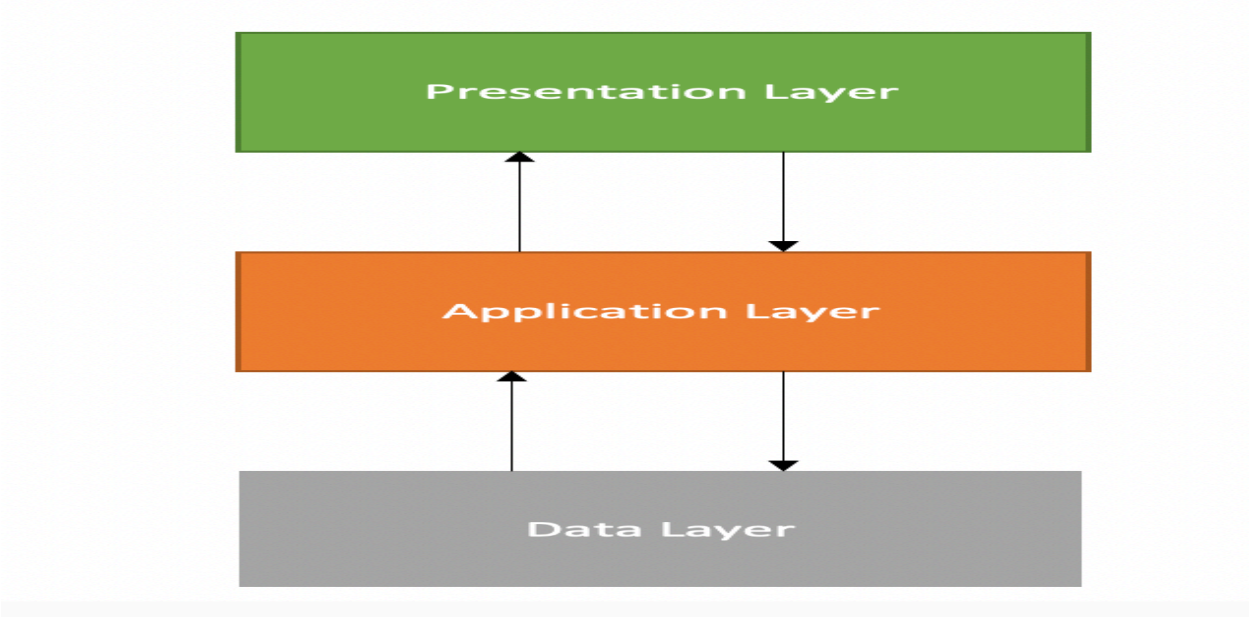


Fig 3: System Architecture [5]

2.3 System Modules

The system is divided into five well-defined modules, each addressing a specific HR function.

Module One: User Management

This module allows for tracking and accessing users by managing accounts, logins, profile edits, and the Employees and HR roles, as well as providing a way for HR to view the list of employees who are currently employed by the organisation. In addition, this module verifies that only the correct people can access the functionality they should through secure access and use of their username, password, and other credentials.

Module Two: Leave Management

This module automates the leave requests process by enabling users to submit a leave request with a reason for the request, time, and type of leave requested, along with the employee's HR policy in place for processing the application, and allowing HR to approve or deny the leave request. Using this module, employees will no longer need to

submit a request for leave on paper, nor will there be any ambiguity or misunderstandings surrounding the leave request between employees and HR.

Module Three: Attendance Management

This module allows HR to track employees' attendance on a daily basis, displays the attendance tracking for employees with month-to-month tracking, enables employees to view their own daily attendance records, and provides HR with retention information for deducting from their payroll checks for absences. This module will assist HR in monitoring employees' attendance and help ensure that payroll is being processed accurately.

Module Four: Salary & Payroll Module

Handles the calculating and disbursing of pay, such as salary allocations to your company, holding employee type allowances, deductions, and base pay, calculating employee salary payments, providing employees a breakdown of the sum of salary and deductions, and maintaining an employee salary history. This allows your company to easily and accurately disperse salaries, as well as ensuring that you pay your employees timely.

Module Five: Department & Deduction Module

Support module which provides a means to manage department structures, including how to set up departments in your organization, maintain data on which employee is in what department, and hold monthly absences and deduct from payroll.

2.4 Security and Verification

Security is an essential requirement, as the system deals with sensitive personal and financial employee data.

1. Authentication

The system ensures that every User must register and log in to the system using a username and a password. The passwords are also stored securely by hashing them.

2. Authorization

The system ensures that only registered users can access the system, and employees will not be able to access the admin dashboard. Employees can only view their own information. But the HR can access the employees' information

3. Input Validation

The system validates the input of the systems, such as the email format, date format, salary data type, and mandatory fields must be filled validations for all forms and available user inputs.

4. Data Security

The system prevents unwanted access to payroll and attendance details. The system also handles the sensitive financial data safely.

5. Session Management

The system uses automatic logout after inactivity and also prevents session misuse.

6. Database Integrity

The system prevents accidental data loss through constraints and validation rules.

2.5 Software Development Tools

Category	Tools/Technology	Description
Frontend	HTML	Used for structuring web pages and interface content.
	CSS	Provides styling, layout, and visual formatting for web pages.
	JavaScript	Adds interactivity and dynamic behavior to the frontend.
Backend	PHP	Handles application logic, form processing, security, and database communication.
Database	MySQL	Stores employee records, applications, admin actions, and system data in a structured manner.
Server	XAMPP	Local server package used for development, hosting, and testing of the web application.
Code Editor	VS Code	Used to write and maintain source code.
Version Control	Git	Helps maintain code versions and track development progress

2.6 Advantages of Using The System

1. Paperwork can be eliminated, and the amount of time to manually process paperwork can be decreased.
2. There will be decreased human error with respect to attendance, leave, and salary calculations.
3. Real-time access to HR information is available to all employees.
4. Data provided will assist in accurate decision-making regarding organizational policies.
5. Employee-HR communications will be more transparent due to improved data available to both parties.
6. Records will be securely stored and the information easily retrievable.
7. Employees in HR will be able to manage many different roles and responsibilities from one location.
8. The scalable system design allows for future improvements such as biometric login, notifications, and analytics.
9. Through improved productivity, this system improves Operational Efficiency.
10. This system will be beneficial to both Small and Large Organizations.

System Life Cycle

The HR Management System is developed using the **Iterative Waterfall SDLC Model**. This model is sequential and easy to manage, making it suitable for academic and structured organizational projects.

3.1 Software Development Life Cycle

SDLC is a step-by-step approach that ensures the system is developed systematically and meets all requirements.

3.1.1 SDLC Phases

1. **Requirement Analysis**

In this phase, the requirements are being collected by documenting and observations. The requirements were analyzed, and the fixed requirements were documented to implement.

2. **System Design**

After the requirement analysis, the design of the System was done by creating the necessary diagrams that explain the whole system's operations.

3. **Implementation**

In this section, the system has been developed using HTML, CSS, JavaScript, and for the backend, PHP has been used.

4. **Testing**

After Implementation, the system has been used many times and tested by giving multiple inputs to check if the system works perfectly or not.

5. **Deployment**

The system has been deployed in a Local server using XAMPP to get client and user feedback and improve further.

6. **Maintenance**

If any issues occurred after deployment, those were analyzed and fixed in the maintenance phase.

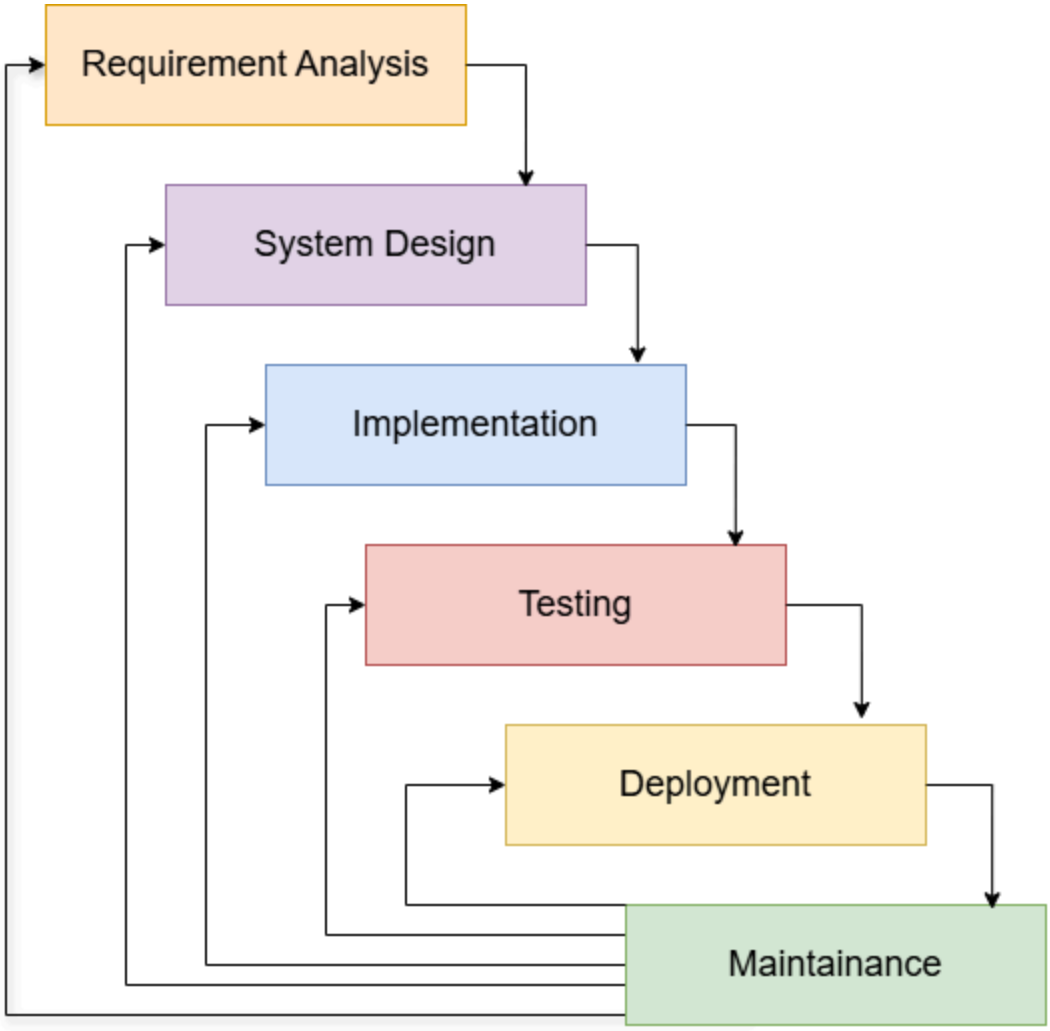


Fig 4: SDLC Process Diagram

Requirement Engineering

Requirement Engineering ensures the system requirements are collected, refined, verified, and documented correctly.

4.1 Requirement Elicitation

During the Requirement Elicitation phase, various methods like Observation and Documentation have been used. The traditional documents of the HR system have been reviewed to gather requirements. By observing the HR workflows, some features have been found. Some common features of HR have been studied. By following these processes, the requirements of the HR Management System have been gathered.

4.2 Requirement Analysis

4.2.1 Feasibility Study

Feasibility Type	Explanation
Operational Feasibility	The system supports essential HR operations such as leave, attendance, and payroll. The interface is easy for both HR and employees
Technical Feasibility	The system can be built using HTML, CSS, JavaScript, PHP, and MySQL, which are widely available and cost-effective.
Economic Feasibility	The project has a low development cost, and the maintenance cost is minimal

Schedule Feasibility	The system can be completed within the academic semester timeline

4.2.2 Requirement Classification

All requirements were classified into two classes, which were Functional Requirements and Non-Functional Requirements.

Functional Requirements

1. User authentication
2. Apply leave
3. Approve/reject leave
4. Record attendance
5. View attendance
6. Allocate salary
7. Pay salary
8. View salary

Non-Functional Requirements

1. System performance
2. Security
3. Usability
4. Reliability
5. Scalability
6. Maintainability
7. Availability

4.2.3 Conflict and Redundancy Resolution

Conflicts between user roles were resolved through role-based access control. Duplicate requirements were merged or removed. Field validation rules were applied to

all modules. User interface conflicts were resolved for consistency.

4.3 Requirement Documentation

The documented requirements include:

1. All functional and non-functional requirements
2. Use case diagrams and narrations
3. ERD, DFD, and Class Diagram
4. CRUD matrix
5. Feature descriptions
6. Security and system architecture
7. Risk analysis and Gantt chart

This documentation ensures the system can be implemented correctly without ambiguity.

Functional requirements

An HR system will allow employees to register an account; employees' accounts will allow employees to log in via the internet and edit/update any information entered on their account. All records of employees will be visible to HR. Employees will be able to submit requests for leave, and HR will review and approve or deny leave requests. HR will track the attendance of employees as well as provide the employee with a complete history of their attendance. The system must be responsible for the calculation, payment, and reporting of salaries to employees; therefore, HR will have the ability to assign salary rates, pay salaries, and provide the employee with a detailed report of their salary and pay status. Additionally, the HR system will provide employees with a means of logging into the system securely, will provide employees with the ability to choose the type of access into the system they want based upon their role, and will validate all entries made by employees.

5.1 Feature List with Description

Feature Name	Description
Account Creation	Allows a new employee to create a system account by entering personal information, contact details, and a password. Validations ensure correctness and prevent duplicate accounts.
User Login	Enables employees and HR to securely log into the system using email and password. The system verifies credentials and redirects users based on role.
Profile Editing	Provides employees with the ability to update their personal profile information. Ensures that stored data remains accurate and up-to-date.

Employee List (HR)	Displays a complete list of all registered employees to the HR user. Includes essential details for supervision and analysis.
Apply Leave	Allows employees to submit leave requests by specifying leave type, duration, and reason. The system validates all entries before submission.
Leave Status Tracking	Enables employees to check the status of their submitted leave applications, including pending, approved, or rejected
Record Attendance (HR)	HR records the daily attendance of employees, marking them as present or absent. Data is stored for payroll processing
View Attendance (Employee)	Employees can view their attendance summary for a selected time period. Helps maintain transparency.
Attendance History (HR)	HR can review the attendance of all employees to monitor discipline and generate internal assessments.
Salary Allocation (HR)	HR defines monthly salary components for each employee, including basic salary, allowance, and deductions.
Pay Salary (HR)	HR processes salary payments for employees. Payment status is updated in the system for monitoring.
View Salary Details (Employee)	Employees view a salary breakdown showing gross salary, deductions, and net payable amount.
Salary Status	Employees view whether their monthly salary is paid or pending, improving financial transparency
Role-Based Access	Ensures employees and HR users view only the information permitted to their roles, enhancing system security.

Data Validation	All input fields undergo validation checks for correct formats and required fields before data is submitted
Secure Data Storage	The system ensures all sensitive data, such as salary records, attendance logs, and personal details, are stored securely.

5.2 Feature Table

Feature	Sub-Feature	Description / Task	Validation Requirements	Actor
Authentication	Create Account	Employee creates an account with personal details	All required fields, a unique email, and a valid password	Employee
	Login	User logs in using email & password	Credential match; role validation	Employee, HR
User Profile Management	Edit Information	Employee updates personal info	Valid data format; fields cannot be blank	Employee
	View Employee List	HR views the complete list of employees	Must be logged in as HR	HR
Leave Management	Apply Leave	Employee submits leave request	Leave type, reason, start–end dates required	Employee

	View Application Status	Employee sees approved/rejected/pending status	Existing leave applications must exist	Employee
	View Leave Applications	HR views all leave submissions	User must be HR	HR
	Update Application Status	HR approves or rejects the leave request	HR must provide the rejection reason if rejected	HR
Attendance Management	View Attendance Records	Employee views the attendance summary	Must be logged in	Employee
	Record Attendance	HR marks present/absent status	Employee must exist; valid date	HR
	View Attendance History	HR sees employee attendance logs	HR authentication required	HR
Payroll Management	View Salary Details	Employee views net salary, breakdown	Salary record must exist	Employee
	View Salary Status	Employee checks if salary is paid	Valid salary payment record	Employee

	Allocate Salary	HR assigns monthly salary components	Valid numerical values; employee exists	HR
	Pay Salary	HR processes salary payment	Salary must be allocated first	HR

System Specification Diagrams

6.1 Entity Relationship Diagram

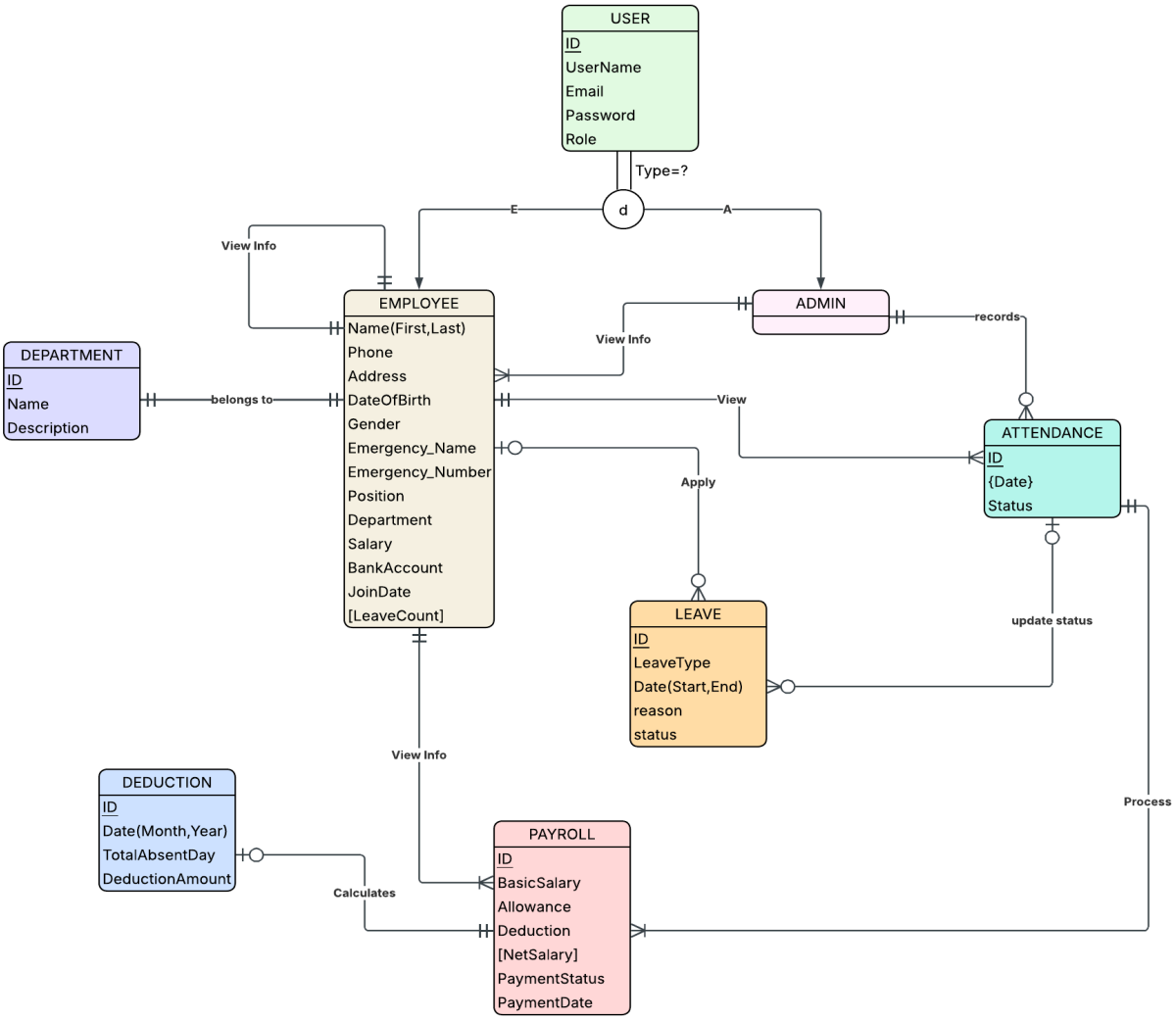


Fig 5: ER Diagram

6.3 Data Flow Diagram

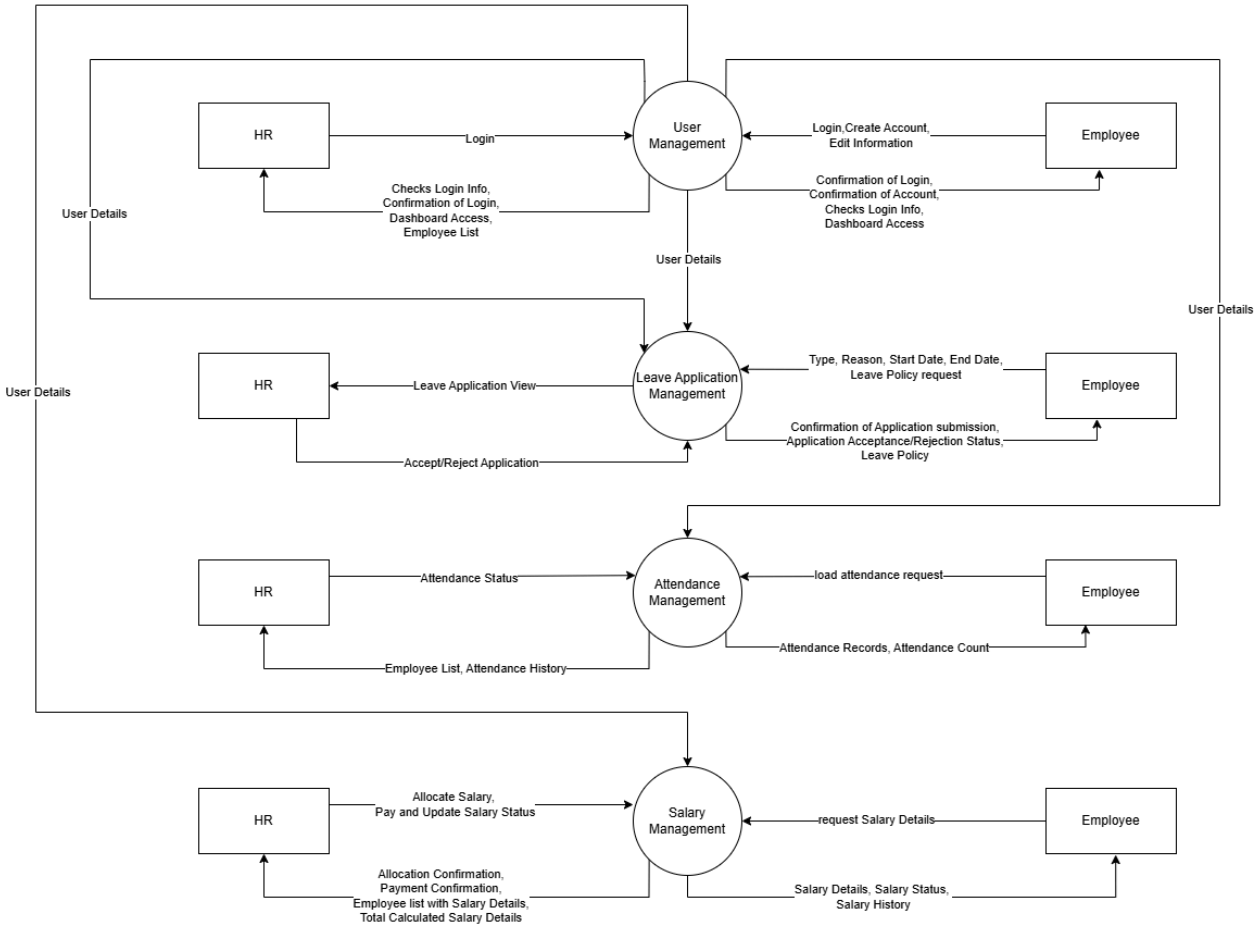


Fig 6: Data Flow Diagram

Link:

https://drive.google.com/file/d/1aCxTv5cDMvRzLGGTn9fYgjkDcJs_JkpA/view?usp=sharing

6.4 Use Case Diagram

6.4.1 User Management Subsystem

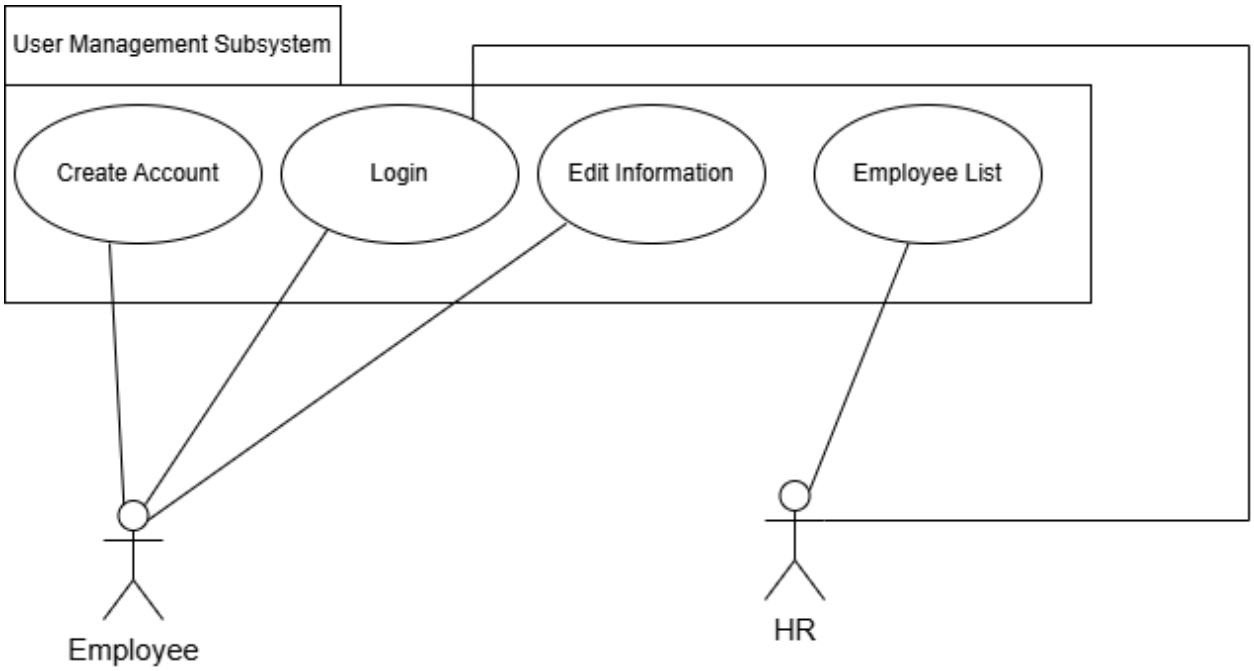


Fig 7: User Management Subsystem Use Case Diagram

6.4.2 Leave Application Subsystem

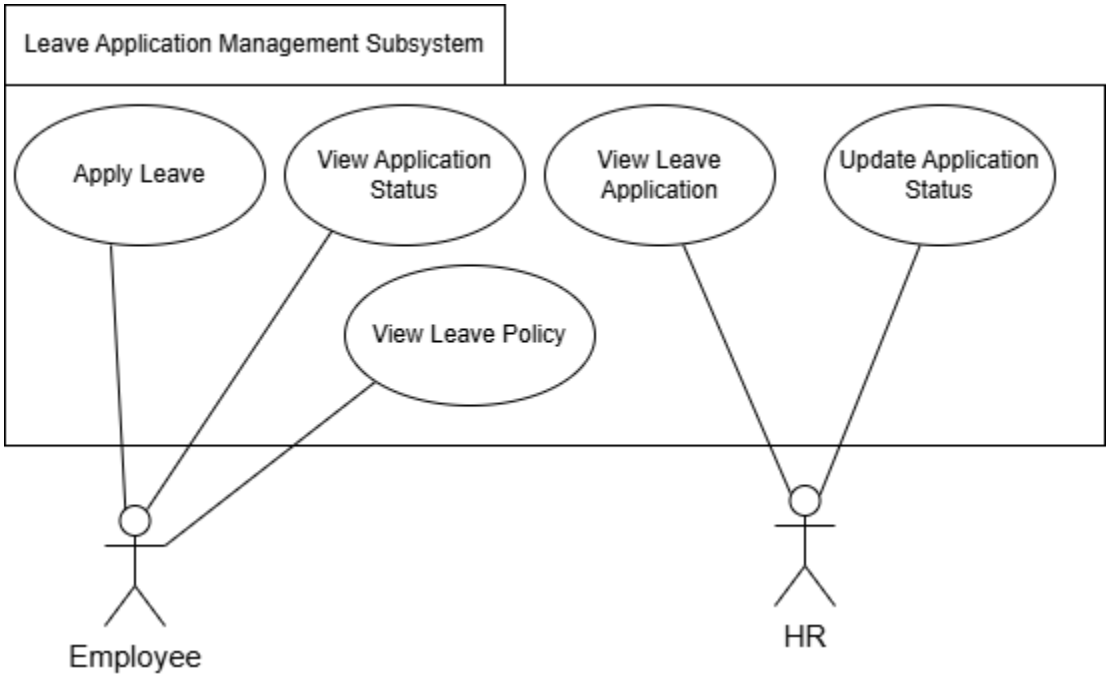


Fig 8: Leave Application Management Subsystem Use Case Diagram

6.4.3 Attendance Management Subsystem

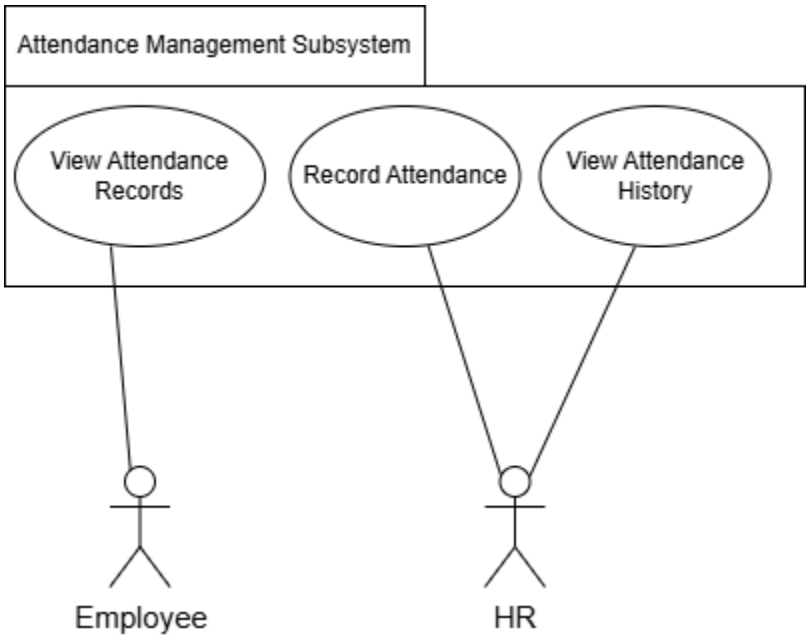


Fig 9: Attendance Management Subsystem Use Case Diagram

6.4.4 Salary Management Subsystem

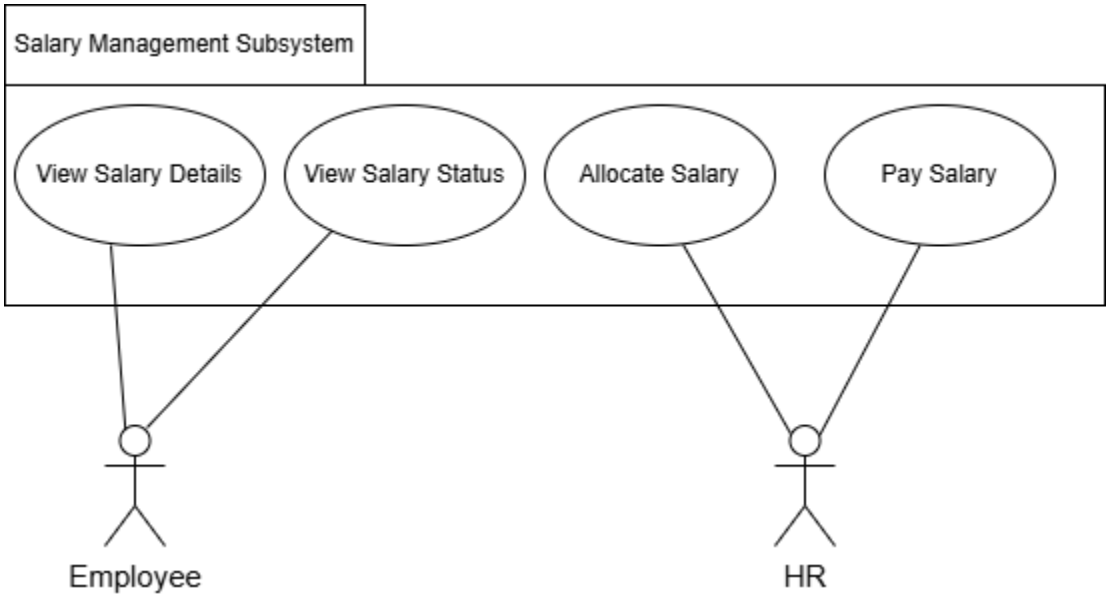


Fig 10: Salary Management Subsystem Use Case Diagram

6.5 Use Case Narration Table

6.5.1 User Management Subsystem Use Case Narration Table

Use Case Name	Create Account	
Actors	Employee	
Description	The employee creates a new account by providing the required personal and login details.	
Precondition	The employee does not already have an account. All required information must be valid.	
Typical course of events	**Actors Events**	**System Response**
	1. Employee enters personal and login information.	2. System validates user input.
	3. Employee clicks "Create Account".	4. System stores account information in the Database and confirms creation.
Alternative	Alt 02: system displays "User already exists". Alt 04: system requests all necessary Data to be filled.	
Post Condition	An employee can log in to the system using the username and Password	

Use Case Name	Login	
Actors	Employee, HR	
Description	Users access the system based on their assigned roles.	
Precondition	Valid accounts must exist.	
Typical course of events	**Actors Events**	**System Response**
	1. The user enters their email and password.	2. System verifies credentials.
	3. User clicks "Login".	4. System loads the dashboard based on role.
Alternative	Alt 02: System displays Invalid User name or password. Alt 04: system notifies that no account exists and asks for registration.	
Post Condition	The user can access all the Features of the system	

<table>
<tr><td>Use Case Name</td><td colspan="2">Edit Information</td></tr>
<tr><td>Actors</td><td colspan="2">Employee</td></tr>
<tr><td>Description</td><td colspan="2">Employee updates personal profile details.</td></tr>
<tr><td>Precondition</td><td colspan="2">Employee must be logged in.
The employee must fill out the Employee Information form.</td></tr>
<tr><td rowspan="4">Typical course of events</td><td>Actors Events</td><td>System Response</td></tr>
<tr><td>1. Employee selects “Edit Profile”.</td><td>2. The system loads the current profile.</td></tr>
<tr><td>3. Employee updates fields.</td><td>4. System validates updated info.</td></tr>
<tr><td>5. Employee clicks “Save”.</td><td>6. System stores updated data in the database.</td></tr>
<tr><td>Alternative</td><td colspan="2">Alt 02: Notifies to fill out the Employee information form

Alt 04: System shows validation error.

Alt 06: System notifies the User of Unable to save information</td></tr>
<tr><td>Post Condition</td><td colspan="2">None</td></tr>
</table>

Use Case Name	View Employee List	
Actors	HR	
Description	HR views the list of all employees.	
Precondition	HR must be logged in.	
Typical course of events	**Actors Events**	**System Response**
	1. HR navigates to the Employee List screen.	2. System fetches and displays all employee profiles.
Alternative	Alt 02: System notifies that no employee is registered	
Post Condition	None	

6.5.2 Leave Application Management Subsystem Use Case Narration Table

Use Case Name	Apply Leave	
Actors	Employee	
Description	Employee submits a leave request.	
Precondition	Employee must be logged in.	
Typical course of events	**Actors Events**	**System Response**
	1. Employee opens “Apply Leave”.	2. System displays leave form.
	3. Employee selects type, dates, and reason.	4. System validates entries.
	5. Employee submits.	6. System stores the request and marks the status “Pending”.
Alternative	Alt 04: System notifies to input valid entries Alt 06: System notifies to fill in all necessary information	
Post Condition	Employee can keep track of their Application status	

Use Case Name	View Application Status	
Actors	Employee	
Description	Employee views the status of previously submitted leave applications.	
Precondition	Must have pending or processed leave requests.	
Typical course of events	**Actors Events**	**System Response**
	1. Employee selects “Leave Status”.	2. System retrieves and displays leave records.
Alternative	Alt 02: System returns that no Application has been made	
Post Condition	Employee can see their apdated application status	

Use Case Name	View Leave Applications	
Actors	HR	
Description	HR views all submitted leave applications.	
Precondition	HR must be logged into the system.	
Typical course of events	**Actors Events**	**System Response**
	1. HR opens the Leave Application panel.	2. System displays all pending and processed requests.
Alternative	Alt 02: System shows No application has been made	
Post Condition	HR can update the Application status	

6.5.3 Attendance Management Subsystem Use Case Narration Table

Use Case Name	Record Attendance	
Actors	HR	
Description	HR marks the daily attendance of employees.	
Precondition	Employee records must exist.	
Typical course of events	**Actors Events**	**System Response**
	1. HR selects the date and employee.	2. System loads the attendance panel.
	3. HR marks Present/Absent.	4. System stores attendance.
Alternative	Alt 02: System notifies that an error occurred Alt 04: System notifies to insure the current date	
Post Condition	HR can see Attendance history	

Use Case Name	View Attendance Records	
Actors	Employee	
Description	Employee views own attendance history.	
Precondition	Attendance records must exist.	
Typical course of events	**Actors Events**	**System Response**
	1. Employee opens “My Attendance”.	2. The system shows attendance details.
Alternative	Alt 02: System notifies that no record has been made	
Post Condition	None	

Use Case Name	View Attendance History	
Actors	HR	
Description	HR views complete attendance logs for all employees.	
Precondition	HR must be logged into the system.	
Typical course of events	**Actors Events**	**System Response**
	1. HR requests attendance history.	2. System retrieves and displays logs.
Alternative	Alt 02: System notifies that no record has been made	
Post Condition	None	

Use Case Name	Update Application Status (Approve/Reject)	
Actors	HR	
Description	HR approves or rejects employee leave requests.	
Precondition	A pending leave request must exist.	
Typical course of events	**Actors Events**	**System Response**
	1. HR selects a leave request.	2. System loads request details.
	3. HR chooses Approve or Reject.	4. System updates status.
Alternative	Alt 04: If rejected then the system asks to give reason	
Post Condition	Employee can see updated leave application status	

6.5.4 Salary Management Subsystem Use Case Narration Table

Use Case Name	View Salary Details	
Actors	Employee	
Description	Employee views monthly salary details, including allowances, deductions, and net salary.	
Precondition	Payroll record must exist.	
Typical course of events	**Actors Events**	**System Response**
	1. Employee opens “Salary Details”.	2. System loads salary information.
Alternative	Alt 02: System notifies Employee that no information is available	
Post Condition	An employee can update Salary information	

Use Case Name	View Salary Status	
Actors	Employee	
Description	Employee checks payment status (Paid/Unpaid).	
Precondition	The employee must log in to the system	
Typical course of events	**Actors Events**	**System Response**
	1. Employee selects “Salary Status”.	2. System loads payment status.
Alternative	Alt 02: System notifies that no payment has been made	

Post Condition	Employee can see the updated status

Use Case Name	Allocate Salary	
Actors	HR	
Description	HR allocates a monthly salary for each employee.	
Precondition	Employee must exist.	
Typical course of events	**Actors Events**	**System Response**
	1. HR enters salary components.	2. System validates all fields.
	3. HR confirms allocation.	4. System stores payroll data.
Alternative	Alt 02: System notifies to input valid data Alt 04: System notifies unable to store data message	
Post Condition	Employees can view Salary information	

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Use Case Name	Pay Salary	
Actors	HR	
Description	HR processes salary payments for employees.	
Precondition	Salary should be allocated.	
Typical course of events	**Actors Events**	**System Response**
	1. HR selects an employee.	2. System loads salary record.
	3. HR clicks “Pay Salary”.	4. System updates payment status and confirms.
Alternative	Alt 02: System notifies: No salary records are available Alt 04: System notifies that some errors occurred	
Post Condition	Employees can see the updated salary payment status	

6.6 CRUD Matrix

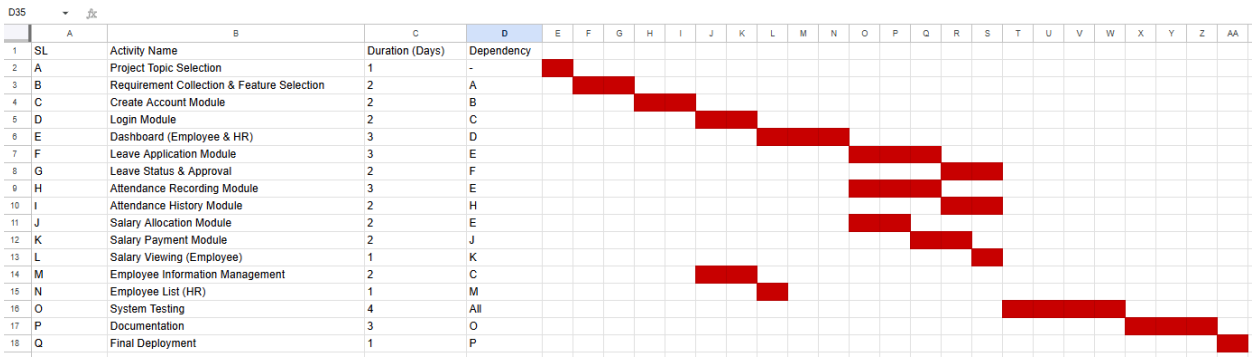
Process/Entity	Employee	HR
User Account Employee	CR	R
Employee Profile	RU	R
Leave Application	CR	RU
Attendance Records	R	CRU
Salary Details (Payroll)	CR	RU
Department Data	R	R
Deduction Data	R	CRU
Payment Status	R	C
Employee List		RD

6.7 Activity Table

SL	Activity Name	Duration (Days)	Dependency
A	Project Topic Selection	1	-
B	Requirement Collection & Feature Selection	2	A
C	Create Account Module	2	B
D	Login Module	2	C
E	Dashboard (Employee & HR)	3	D
F	Leave Application Module	3	E
G	Leave Status & Approval	2	F
H	Attendance Recording Module	3	E
I	Attendance History Module	2	H
J	Salary Allocation Module	2	E
K	Salary Payment Module	2	J

L	Salary Viewing (Employee)	1	K
M	Employee Information Management	2	C
N	Employee List (HR)	1	M
O	System Testing	4	All
P	Documentation	3	O
Q	Final Deployment	1	P

6.8 Gantt chart



SL	Activity Name	Duration (Days)	Dependency
A	Project Topic Selection	1	-
B	Requirement Collection & Feature Selection	2	A
C	Create Account Module	2	B
D	Login Module	2	C
E	Dashboard (Employee & HR)	3	D
F	Leave Application Module	3	E
G	Leave Status & Approval	2	F
H	Attendance Recording Module	3	E
I	Attendance History Module	2	H
J	Salary Allocation Module	2	E
K	Salary Payment Module	2	J
L	Salary Viewing (Employee)	1	K
M	Employee Information Management	2	C
N	Employee List (HR)	1	M
O	System Testing	4	All
P	Documentation	3	O
Q	Final Deployment	1	P

Fig 11: Gantt Chart

Link:

https://docs.google.com/spreadsheets/d/1B--hmHQKbjIXtAzNNQEz9tBjQqAlwuudSb7wnwfTNto/edit?usp=sharing

Non-Functional Requirements

The non-functional requirements define the overall quality attributes of the HR Management System. They ensure that the system performs efficiently, remains reliable, and supports long-term scalability. The following subsections describe each requirement in detail.

7.1 Feature List with Description

Usability

The system is designed to be user-friendly and allows both HR and employees to perform tasks. A clean layout, clear labeling, and consistent interface design help users navigate the system easily. For example, an employee can create an account in fewer than five clicks.

Performance

The system must operate efficiently even under high usage. All pages should load within three seconds. The backend must handle multiple concurrent requests like, attendance submissions, or leave applications and without slowing down or crashing.

Security

All passwords are stored using secure hashing algorithms to prevent exposure in case of database breaches. Employees and HR users have different permission levels. Employees cannot access HR dashboards or modify data they do not own. All user inputs are validated to prevent SQL, incorrect formats, and unauthorized access attempts.

Reliability

The system must function consistently without unexpected shutdowns or data loss. Critical operations like salary processing or attendance recording must work reliably

every time. Data stored in the system should always remain accessible and correct, even after power failures or sudden interruptions.

Scalability

The HR Management System is designed to accommodate organizational growth. As the number of employees increases, the system must continue to perform efficiently. for example: biometric attendance integration, automated notifications, analytics dashboards, and multiple department support.

Maintainability

The system's codebase is modular, meaning each feature is separated into manageable components. This structure allows developers to fix issues or update functionalities without affecting other modules. All code is properly commented, organized, and documented, making it easier for future developers to maintain or extend the system.

Portability

The system is platform-independent and can run on various devices and operating systems. It supports all major web browsers such as Chrome, Firefox, Edge, and Safari. Additionally, the interface is responsive, allowing access from desktops, laptops, tablets, and mobile devices. This ensures that employees and HR personnel can use the system anytime and from any compatible device without loss of functionality.

Data Integrity

The system ensures that all stored data remains accurate, consistent, and complete. Validation rules prevent incorrect or incomplete information from being entered into the database. For example, dates cannot be in an invalid format, salary numbers must be valid numerical values, and duplicate employee accounts are not allowed.

Result Analysis

8.1 SWOT Analysis

Strength	Weakness
Reduces manual HR work	Dependent on the internet
Transparent leave & payroll system	Requires training for HR
Accurate attendance tracking	Limited to only HR and the Employee
Opportunities	**Threats**
Can integrate biometric attendance	Cybersecurity risks
Possible mobile app expansion	Server failures

8.2 Risk Matrix

Risk Factor	Likelihood	Impact	Solution
Data Loss	Low	High	Automated backups
Unauthorized Access	Medium	High	Role-based login
Incorrect Payroll	Medium	Medium	Automated calculations
Server Failure	Low	High	Cloud hosting/redundancy

Conclusion

This report proposes a comprehensive and systematic approach to automating HR Management System activities. The proposed HR Management System solution includes all critical HR activities, including but not limited to: employee management, processing of leave requests, Attendance model, and payroll Management systems. All employees will be able to view their own information and view their pay and leave dates on a single, web-based platform.

The designed HR Management System has improved accuracy, enhanced transparency for employees accessing their information, and increased company productivity by minimizing duplicate administrative tasks. In addition to the benefits of accuracy, transparency, and productivity, the security features protect sensitive employee information.

In summary, the HR Management System is designed to provide a scalable, secure, and practical toolset for improving the effectiveness and efficiency of HR practices in contemporary organizations.

References

1. Lingareddy, S. R. (2007). *The Design and Implementation of Human Resource Management Website* (Doctoral dissertation, Indiana University South Bend).

2. Bannikov, S. A., & Abzeldinova, K. T. (2021, February). Digital transformation of the HR management system. In *International Scientific and Practical Conference "Russia 2020-a new reality: economy and society"(ISPCR 2020)* (pp. 54-58). Atlantis Press.

3. Navaz, A. S., Fiaz, A. S., Prabhadevi, C., Sangeetha, V., & Gopalakrishnan, S. (2013). Human Resource Management System. *arXiv preprint arXiv:1309.5351*.

4. Danyliuk, I., Babala, L., & Khoma, N. (2022). HRM systems of personnel management. *Ekonomichnyy analiz*, *32*(3), 240-246.

5. Yap Wen Jun Bryan. (2020, October 25). *Software Architecture Patterns — Layered Architecture Patterns*. Medium. Retrieved December 7, 2025, from

https://yapbryan6.medium.com/software-architecture-patterns-layered-architecture-patterns-975ff2abec6f